NCAA March Madness Prediction System

Project Update #3: Breaking Through the Ordinal Ceiling

Roddy Huang, Yumeng Jin
Cornell University

March 29, 2026

1 Overview

Update #2 ended with a puzzle. We had implemented eight models spanning three tiers of complexity, and the best one was a logistic regression on a single ordinal rank difference, at 0.196 Brier. Everything else—XGBoost on 53 features, Mixture-of-Experts, a 4-layer Transformer—was worse. The report identified the likely culprit: the Massey ordinals file only provides integer ranks, and discretizing a continuous efficiency margin into 363 bins throws away most of the signal.

This update is about what happened when we fixed that. We built a vectorized opponent-adjusted efficiency engine from the 124K regular-season box scores, producing continuous AdjOE/AdjDE/AdjEM values that correlate at $r = 0.99$ (Spearman) with KenPom's official rankings. We then combined this with the Kellert spread-to-probability conversion [Kellert, 2017]—predicting continuous point margins instead of binary outcomes, then converting via the Normal CDF. The result: **0.1878 Brier**, a 0.008 improvement that moves us past the KenPom logistic benchmark (∼0.185) and into the range where Kaggle competition placements start to matter.

We also ran four additional experiments—Mixture-of-Experts by game type, isotonic/Platt calibration, 2021 COVID season exclusion, and market-prior fusion—to squeeze further improvement. None of them worked. The reasons why are instructive.

2 Self-Computed Efficiency Ratings

2.1 Motivation

The gap between ordinal rank and continuous efficiency was the single largest information loss in our pipeline. A rank difference of 4 (e.g., #1 vs #5) could correspond to an AdjEM gap of 15 points or 2 points—we had no way to tell. KenPom's actual AdjEM values are behind a $25 paywall and Barttorvik's website employs JavaScript anti-scraping protections. So we built our own.

2.2 Method

The algorithm replicates the core of the KenPom/Barttorvik methodology in three steps:

1. **Per-game efficiency.** For each game, estimate possessions as $\text{Poss} = \text{FGA} - \text{OR} + \text{TO} + 0.44 \cdot \text{FTA}$, then compute per-100-possession offensive and defensive efficiency.
2. **Season averages.** Aggregate per team using possession-weighted means.
3. **Iterative opponent adjustment.** For each game, scale the team's OE by the factor $\bar{E}_{\text{league}}/\text{AdjDE}_{\text{opponent}}$. If the opponent has a strong defense (low AdjDE), the denominator is small, boosting the team's adjusted offense. Symmetrically for defense. Repeat until convergence (< 0.01 per iteration, typically 15–20 rounds).

The implementation is fully vectorized using NumPy (no Python loops in the inner iteration) and processes all 24 seasons in under 30 seconds. Results are cached to CSV for reuse.

2.3 Validation

We validated the self-computed ratings against the official KenPom rankings from the Massey ordinals file (2025 season):

- Spearman rank correlation: $r = 0.9915$ ($p < 10^{-300}$)
- Pearson correlation (AdjEM vs ordinal rank): $r = -0.963$
- 9 of the top 10 teams match between our AdjEM ranking and KenPom's
- AdjEM range: -51.7 to $+44.9$ (vs KenPom's typical -25 to $+35$; our range is wider because we lack home-court and tempo adjustments that KenPom uses to regularize)

The near-perfect Spearman correlation means our *ranking* essentially reproduces KenPom's. The wider AdjEM range means our absolute values are noisier, but for matchup prediction we only need the *difference*, which is robust to global scaling.

3 Kellert Spread Model

Update #2's models all predicted $P(\text{win})$ directly. Kellert's 2017 second-place Kaggle solution [Kellert, 2017] instead predicts the continuous point margin $\hat{\Delta}$ via Ridge regression, then converts:

$$P(\text{Team A wins}) = \Phi\left(\frac{\hat{\Delta}}{\sigma}\right) \tag{1}$$

where σ is calibrated from training residuals. The intuition: basketball scores are approximately Normal conditional on team quality, so modeling the full margin distribution produces better-calibrated probabilities than a logistic sigmoid.

Our implementation uses Ridge ($\alpha = 1.0$) on two features: AdjEM difference and seed difference. The flip-and-double augmentation (each game generates two training rows with negated features and inverted outcomes) ensures the model intercept is exactly zero. Probabilities are clipped to $[0.025, 0.975]$.

The calibrated σ averages 11.55 across LOTO folds, consistent with Kellert's reported 10.55 and the common rule of thumb that NCAA game outcomes carry $\sim$11 points of irreducible noise.

4 Results

Table 1 traces the Brier score improvement through each engineering change, starting from Update #2's best model.

Table 1: Brier score progression. Each row adds one change to the previous row. LOTO over 2014–2025 (11 tournaments, 736 games).

Model Configuration	**Mean Brier**	**Δ vs Update #2**
Update #2 baselines (ordinal ranks)		
KenPom rank logistic	0.1961	—
Multi-feature logistic (16 features)	0.197	+0.001
XGBoost (53 features)	0.214	+0.018
MoE (K=4, EM)	0.209	+0.013
Transformer (4-layer)	0.245	+0.049
Update #3: continuous efficiency		
AdjEM logistic (single feature)	0.1888	-0.0073
AdjEM logistic + flip-and-double	0.1880	-0.0081
Spread model: AdjEM + seed	0.1883	-0.0078
Spread model: AdjEM + seed + flip	**0.1878**	**-0.0083**

Three things stand out. First, **the continuous AdjEM on its own (0.1888) beats every model from Update #2**, including the 16-feature logistic and the 53-feature XGBoost. One well-constructed feature outweighs 52 noisy ones. Second, the spread-to-probability conversion provides a consistent ∼0.001 improvement over logistic regression with the same features. Third, flip-and-double contributes another ∼0.001 by eliminating the directional bias from canonical team ordering.

5 Negative Results

We tried four additional techniques. None improved the score.

5.1 Mixture-of-Experts by game type

We partitioned matchups into three groups by seed gap: chalk ($|\Delta s| \geq 5$, 59% of games), competitive ($2 \leq |\Delta s| \leq 4$, 19%), and toss-up ($|\Delta s| \leq 1$, 22%). The hypothesis was that these game types have different spread–outcome relationships, particularly different σ values.

Table 2: Per-group spread model analysis. Residual σ is nearly identical across game types.

Game Type	N	**Hi-seed win%**	**Avg margin**	**Residual** σ
Chalk ($\lvert\Delta s\rvert \geq 5$)	433	75.8%	+9.4	11.10
Competitive (2–4)	143	67.1%	+4.0	11.64
Toss-up (0–1)	160	38.1%	−2.5	11.91

The data killed the hypothesis. Residual σ varies by less than 1 point across groups (Table 2). Training separate models per group produced 0.1891 Brier (worse than global), and using a global model with group-specific σ produced 0.1879 (tied). The AdjEM difference already encodes the game type: a chalk matchup has AdjEM diff ∼+19, a toss-up has ∼+1. The global linear model adapts to this through the continuous feature, making explicit partitioning redundant.

This result, combined with Update #2's EM-based MoE (0.209), is a strong signal that **MoE does not help for this problem at this data scale**. With only 143 competitive games and 160 toss-ups across 11 seasons, per-group models cannot learn anything the global model does not already capture.

5.2 Probability calibration

We tested three calibration approaches on top of the best spread model:

- **Isotonic regression**: 0.1915 (+0.0036). Overfits to training probabilities; every season got worse.
- **Platt scaling** (logistic on spread): 0.1881 (+0.0003). Marginal degradation.
- σ **multiplier search** (0.90–1.15): best at $\sigma \times 0.95$, Brier = 0.1878. Tied with baseline.

The Normal CDF is already an excellent calibration function for this domain. The spread model's residuals are approximately Normal (verified in Update #2's diagnostics), so re-calibrating adds parameters without adding information.

5.3 2021 COVID season exclusion

2021 is every model's worst year (Brier ∼0.213 vs ∼0.185 for normal years). We tested three treatments: (a) exclude 2021 from training only, (b) exclude entirely, (c) downweight by 50%. All three produced identical results for non-2021 seasons (Brier 0.1853). The 2021 training data does not poison the model; 2021 is simply hard to predict because the COVID-disrupted season created a genuine regime shift.

5.4 Market-prior fusion

Without access to Vegas or Kalshi game-level odds (Kalshi has no NCAA markets; we have no odds API key), we used historical seed-pair win rates as a proxy for "market consensus." Bayesian log-odds fusion with the spread model hurt at every prior weight we tested (0.1 to 0.5). The reason: seed information is already in the model via the `seed_diff` feature. Fusing a seed-based prior with a model that already uses seeds double-counts the same signal.

This technique would likely help with *actual* market odds, which incorporate information our model does not have (injuries, motivation, public betting patterns). But without that data, there is nothing to fuse.

6 Competition Submission

We generated the Kaggle Stage 2 submission for the 2026 tournament using the best model (Ridge spread, AdjEM + seed, flip-and-double, $\sigma = 11.55$, clip $[0.025, 0.975]$). The model was trained on all available tournament games (2014–2025, 1,472 rows after flip-and-double). The submission contains 132,133 rows of P(Team A beats Team B) for every possible D1 matchup. Sample predictions for top seeds align with expectations: Duke (AdjEM = 44.9) gets ∼63% against other top teams and >97% against 16 seeds.

7 What We Learned

One good feature beats many mediocre ones. Our single AdjEM difference (0.1888) beats a 16-feature logistic (0.197), a 53-feature XGBoost (0.214), and a 4-layer Transformer (0.245). In small-data regimes, signal density matters more than feature breadth.

Continuous trumps discrete. The same underlying information (team quality) expressed as a continuous efficiency margin is worth 0.008 Brier over its ordinal rank encoding. This is approximately the gap between a median Kaggle submission and a top-20% one.

The spread model is naturally well-calibrated. Predicting margins and converting via $\Phi(\hat{\Delta}/\sigma)$ produces probabilities that do not need post-hoc calibration. The Normal residual assumption holds well enough for tournament games.

Complexity does not help at 736 games. MoE, Transformer, XGBoost, ensemble optimization, and isotonic calibration all failed or tied. The common thread is insufficient data: each technique adds parameters or partitions the data, and with ∼67 test games per LOTO fold, there is not enough signal to exploit.

Negative results are results. We spent significant time on MoE, calibration, COVID exclusion, and market fusion. None moved the needle. But each "failure" eliminated a hypothesis and narrowed the search space. The MoE analysis (Table 2) shows precisely why game-type partitioning fails: the residual σ is invariant to seed gap. That insight—that tournament game noise is constant regardless of matchup quality—is worth knowing.

8 Current Standing and Remaining Gap

The remaining 0.013 gap to Vegas likely requires information we do not have: injury reports, late-season player developments, travel fatigue, and the aggregated wisdom of the betting market itself. Our model uses only box-score-derived efficiency and committee-assigned seeds. Within that information set, we believe we are close to the ceiling.

Table 3: Where we stand relative to known benchmarks.

Benchmark	Brier	Gap to Our Best
Coin flip	0.250	+0.062
Seed logistic	0.200	+0.012
KenPom rank logistic	0.196	+0.008
Our best (spread + AdjEM)	**0.188**	—
KenPom AdjEM logistic (literature)	~0.185	−0.003
Vegas line	~0.175	−0.013
Kaggle top-10	<0.170	−0.018

9 Next Steps

1. **Acquire real market odds.** An Odds API subscription ($20/month) would give us historical Vegas lines for every tournament game since 2014. This is the single feature most likely to close the remaining gap.
2. **Improve efficiency model.** Add home-court adjustment, recency weighting (games in March count more than November), and margin-of-victory dampening (blowouts are less informative).
3. **Ensemble with Transformer.** Pretrain the Transformer on regular-season outcomes, then fine-tune on tournaments. Even if the Transformer alone is worse, it may capture sequence-dependent patterns (momentum, hot streaks) that the static efficiency model misses, providing diversification in an ensemble.
4. **Final paper.** Consolidate Updates 1–3 into a single paper with complete methodology, ablation study, and competition results.

References

Oliver, D. *Basketball on Paper: Rules and Tools for Performance Analysis.* Potomac Books, 2004.

Kellert, S. March Machine Learning Mania 2017: 2nd Place Solution. Kaggle Winner Interview, 2017.

Landgraf, A. March Machine Learning Mania 2017: 1st Place Solution. Kaggle Winner Interview, 2017.

Pomeroy, K. KenPom.com: College Basketball Ratings. https://kenpom.com, 2004–present.